

Bindu

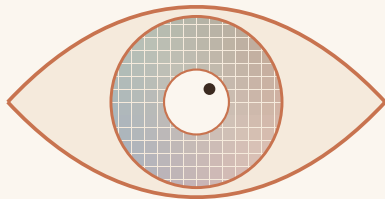
An image workspace built on
two-dimensional signals

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$x[m,n]$ $X[k,l]$

Github: <https://github.com/onigari/bindu>

A picture is just a signal

- Every photo is a grid of samples $x[m, n]$, but photo apps **hide the maths** behind a slider.
- We wanted one workspace where each button is a **signal operation** you can see, measure and undo.
- It is built from the course toolbox: **convolution, the 2D DFT, linear transforms, wavelets** and entropy coding.

QUESTION → TOOL

How are colours spread?	histograms, luminance
Which frequencies?	2D FFT spectra
Blur or sharpen?	convolution kernels
Store it smaller?	Fourier pruning, Haar
Fix an old photo?	morphology, inpainting



System Architecture

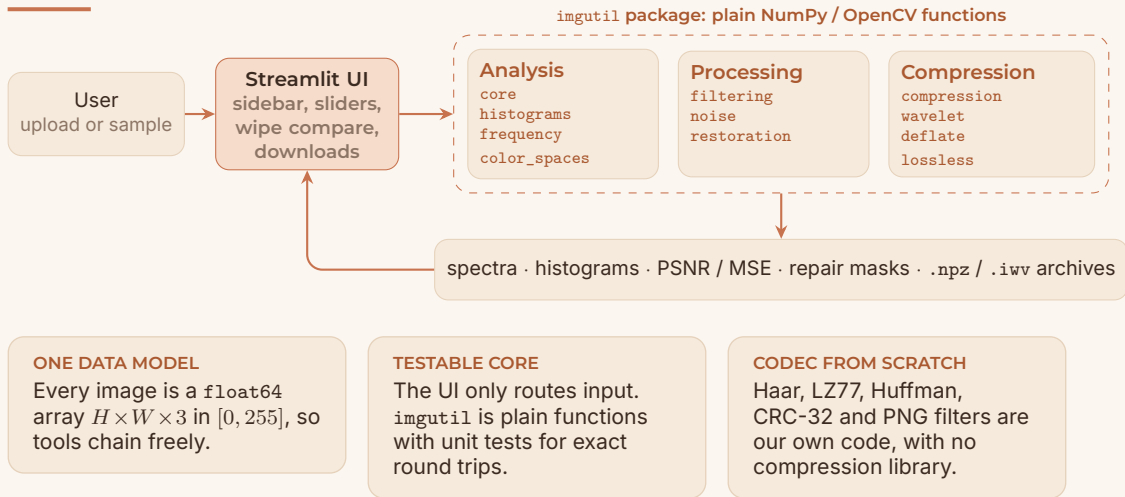
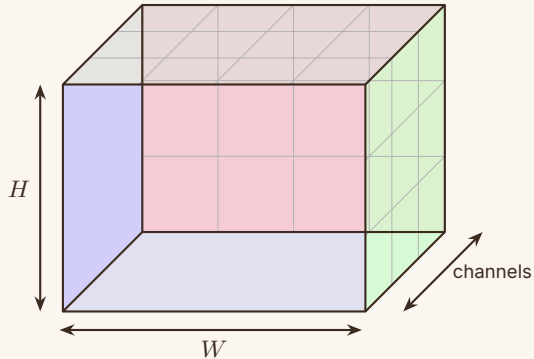
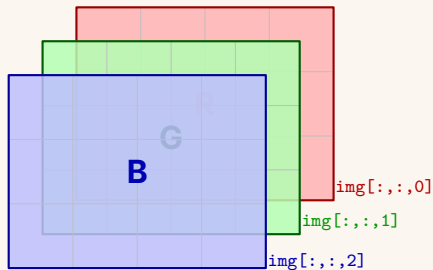


Image as a 3D array

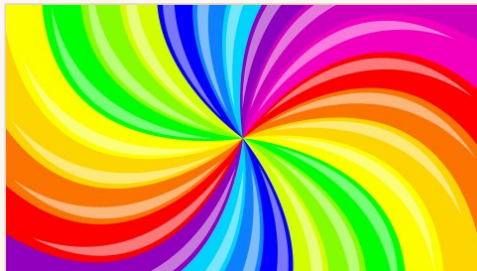
- A colour image is a 3D array of shape $H \times W \times 3$.
- The third axis indexes the three colour channels.



An image is three 2D signals stacked together



$$\begin{aligned} x_c[m, n], \quad c \in \{R, G, B\} \\ 0 \leq m < H, \quad 0 \leq n < W \\ x_c[m, n] \in [0, 255] \end{aligned}$$



Indexing the third axis, `img[:, :, c]`, splits the photo into its R, G and B planes.

The 2D DFT: which frequencies build the picture

2D DISCRETE FOURIER TRANSFORM

$$X[k, l] = \sum_{m=0}^{H-1} \sum_{n=0}^{W-1} x[m, n] e^{-j2\pi(\frac{km}{H} + \frac{ln}{W})}$$

fftshift puts DC in the centre; we display $\log(1 + |X|)$.

PARSEVAL'S THEOREM

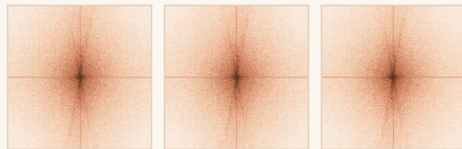
$$\sum_{m,n} |x[m, n]|^2 = \frac{1}{HW} \sum_{k,l} |X[k, l]|^2$$

Energy in space = Energy in frequency.

PARTIAL RECONSTRUCTION

$$\hat{x} = [\mathcal{F}^{-1}X_R, \mathcal{F}^{-1}X_G, 0]$$

LOG-MAGNITUDE SPECTRA, MEAN REMOVED



R

G

B

INVERSE FFT OF CHOSEN CHANNELS



all three

R+G

G+B

Energy crowds at the centre (low frequencies).

Histograms: the statistics of each channel

HISTOGRAM OF CHANNEL c

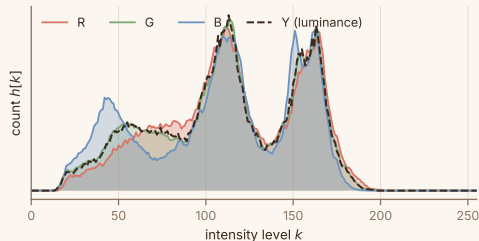
$$h_c[k] = \#\{(m, n) : x_c[m, n] = k\}$$

$$p_c[k] = \frac{h_c[k]}{HW}, \quad \sum_{k=0}^{255} p_c[k] = 1$$

PERCEIVED LUMINANCE

$$Y = \begin{bmatrix} 0.299 & 0.587 & 0.114 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

Y uses the perceived luminance, which is closer to what the eye sees.



Lakeside sample (left). Blue sits highest (sky and water) and red lowest. The dashed curve is Y : narrower, because the channels are averaged inside every pixel.

Colour space: RGB $\rightarrow$ YCbCr is a change of basis

FULL-RANGE BT.601 TRANSFORM

$$\begin{bmatrix} Y \\ C_b \\ C_r \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.169 & -0.331 & 0.500 \\ 0.500 & -0.419 & -0.081 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix}$$

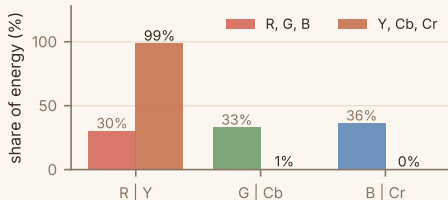
The C_b and C_r rows sum to zero, so a grey pixel ($R = G = B$) gets neutral chroma 128.

WHY IT HELPS

R, G and B move together ($\rho_{RG} = 0.994$ on this image). After the transform almost all the detail is in Y ; C_b and C_r are smooth, so there is less to store.



Y Cb (contrast $\times 5$) Cr (contrast $\times 5$)



Energy about the mean:
split evenly across R, G, B, but 98.9% goes to Y.

Filtering: convolution $\Leftrightarrow$ multiplication

2D CONVOLUTION = LTI SYSTEM

$$y[m, n] = \sum_i \sum_j h[i, j] x[m - i, n - j]$$

$$\Leftrightarrow Y[k, l] = H[k, l] \cdot X[k, l]$$

KERNELS ARE JUST DIFFERENT h

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

box blur: low-pass, $\sum h = 1$

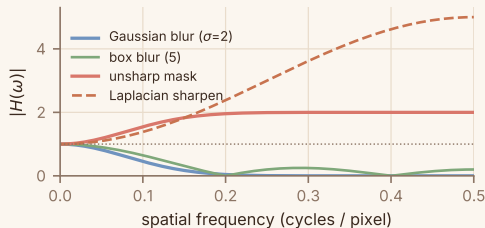
$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix}$$

sharpen: $x - \nabla^2 x$

GAUSSIAN AND UNSHARP MASK

$$G_\sigma \propto e^{-\frac{(i^2+j^2)}{2\sigma^2}}, \quad y = x + a(x - G_\sigma * x)$$

$$\Rightarrow \text{one kernel: } h = (1 + a)\delta - a G_\sigma$$



original

Gaussian, $\sigma=3$

unsharp, $a=1.5$

Linear, so per-channel = whole-image ($\max |\Delta| < 10^{-8}$).

One filter tool, four kernels

Box blur

 5×5

flat average

0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04
0.04	0.04	0.04	0.04	0.04



Gaussian blur

 $25 \times 25, \sigma = 3$

smooth fall-off



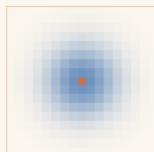
Laplacian sharpen

 $3 \times 3, \text{scale } 0.8$
 $\delta - 0.8 \nabla^2$

-1.6	0	-1.6
0	7.4	0
-1.6	0	-1.6



Unsharp mask

 $17 \times 17, \sigma = 2, a = 1.5$
 $(1 + a) \delta - a G_\sigma$


Weights from `filtering.filter_kernel()`: exact match to the tool, each sums to 1.

■ positive ■ negative

Lossy: keep only the strongest frequencies

REAL IMAGE $\Rightarrow$ CONJUGATE SYMMETRY

$X[k, l] = X^*[-k, -l] \Rightarrow$ store only half.

ENERGY PRUNING, TARGET η

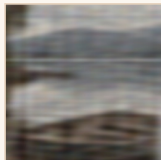
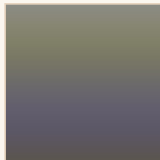
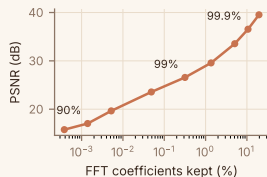
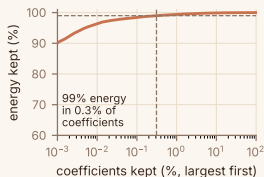
Rank pairs by $E_p = w_p |X_p|^2$; keep the fewest (DC always) with

$$\sum_{p \in S} E_p \geq \eta \sum_p E_p$$

$$\hat{x} = \mathcal{F}^{-1}\{X \cdot \mathbf{1}_S\}$$

MEASURING THE LOSS

$$\text{MSE} = \frac{1}{3HW} \sum (x - \hat{x})^2, \quad \text{PSNR} = 10 \log_{10} \frac{255^2}{\text{MSE}}$$



$\eta=90\% \cdot 17.0 \text{ dB}$

99% $\cdot 26.6 \text{ dB}$

99.9% $\cdot 36.5 \text{ dB}$

Our codec, 512 $\times$ 512 photo: 99% of the energy is in 0.32% of the coefficients. DC counts as energy, so detail returns near 99.9%.

Lossless compression I: integer Haar lifting

REVERSIBLE COLOUR TRANSFORM (INTEGERS ONLY)

$$\begin{bmatrix} R - G \\ G \\ B - G \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix}, \quad \det = 1$$

LIFTING ONE PAIR (a, b)

forward

$$d = b - a$$

$$s = a + \lfloor d/2 \rfloor$$

exact inverse

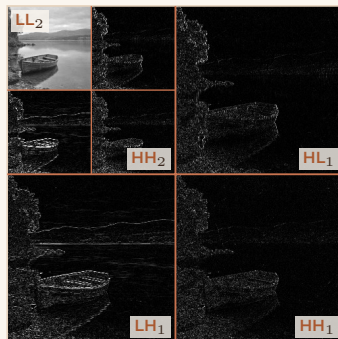
$$a = s - \lfloor d/2 \rfloor$$

$$b = a + d$$

EXAMPLE

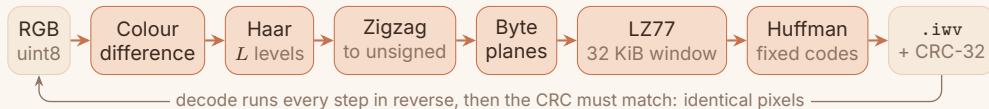
$$(10, 14) \xrightarrow{\text{lift}} (s, d) = (12, 4) \xrightarrow{\text{undo}} (10, 14)$$

Smooth areas give $d \approx 0$, and nothing is rounded away.



Two levels on our sample. Most detail is near zero (dark).

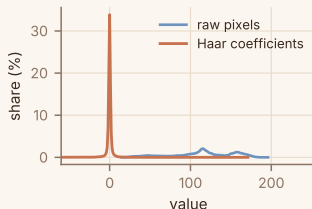
Lossless compression II: coefficients to a .iww file



ZIGZAG MAP

$$z = \begin{cases} 2c, & c \geq 0 \\ -2c - 1, & c < 0 \end{cases}$$

0, -1, 1, -2, 2 $\mapsto$ 0, 1, 2, 3, 4:
small magnitudes become small bytes.



ENTROPY, BITS PER VALUE

$$H = - \sum_v p(v) \log_2 p(v)$$

After the transform, 86% of values lie within ± 2 , so LZ77 and Huffman find far more repetition.

RESULT ON A 256×256 CROP

Entropy (pixels)	7.04 bits
Entropy (coefficients)	2.99 bits
Raw RGB	192.0 KiB
.iww archive	112.0 KiB
Ratio	1.71×
Decoded pixels	bit-exact

Restoration: find the scratches, then fill them

GREY-SCALE MORPHOLOGY

$$f \ominus b = \min_b f \quad \text{erosion}$$

$$f \oplus b = \max_b f \quad \text{dilation}$$

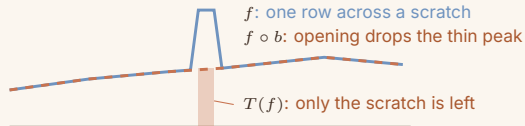
$$f \circ b = (f \ominus b) \oplus b \quad \text{opening}$$

$$T(f) = f - f \circ b \quad \text{white top-hat}$$

MASK, THEN REPAIR

$$M = [T(f) \geq \tau] \oplus (3 \times 3), \quad \tau = 30$$

Telea inpainting rebuilds the masked pixels from their neighbours. A mask over 10% of the image is refused.



damaged

top-hat T mask M

repaired

The mask covers 1.6% of the portrait. Bright hair highlights are picked up too, so a threshold slider controls the trade-off.

Denoising: the right filter depends on the noise

TWO NOISE MODELS

Gaussian: $y = x + n$, $n \sim \mathcal{N}(0, \sigma^2)$

S&P: $y \in \{0, 255\}$ with prob. p , else x

MEDIAN FILTER (NON-LINEAR)

$y[m, n] = \text{median}_{|i|, |j| \leq 2} x[m + i, n + j]$

A lone 0 or 255 is never the middle value, so it is dropped.

NON-LOCAL MEANS

$$\hat{x}(p) = \frac{\sum_q w(p, q) x(q)}{\sum_q w(p, q)}$$

$$w(p, q) = e^{-\frac{\|P_p - P_q\|^2}{h^2}}$$

GAUSSIAN NOISE, $\sigma = 25$



noisy 20.2 dB median 30.0 Gaussian 30.3 **NL-means 31.2**

SALT & PEPPER, $p = 10\%$



noisy 15.8 dB **median 35.8** Gaussian 26.2 NL-means 16.2

PSNR vs. the clean crop (5×5 windows, $h = 15$).

Thank You